Day 070 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Connect Constraints and Solution Sets

Standards: 6.EE.B.8, 6.EE.B.5

Learning target: I can connect Constraints and Solution Sets, and explain and check my reasoning.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 3, 25

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01 / Learn

Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

Worked example 1

Let x be any real number. Write and graph: x is less than 36. Label the endpoint and shade the solution ray.

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. Graph a strict bound with an open circle; use a closed circle for an included endpoint.

3. “less than” means <

4. exclude the endpoint.

5. State the result in the form requested: x < 36; open circle at 36, shade left.

Answer: x < 36; open circle at 36, shade left

Worked example 2

A club needs at least 18 members. Select all true statements for its membership count n.
A. n ≥ 18
B. n > 18
C. 17 is an allowed membership count.
D. 18 is an allowed membership count.

1. Substitute each value and decide whether the comparison is true.

2. At least includes the boundary, so n ≥ 18.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: A, D - n ≥ 18; 18 is an allowed membership count.

Day 070 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Let x be any real number. Write and graph: x is at least 21. Label the endpoint and shade the solution ray.

02 / Constructed response

Which values in {9, 10, 11} satisfy x ≤ 10? Test each by substitution.

03 / Constructed response

Let x be any real number. Write and graph: x is more than 4. Label the endpoint and shade the solution ray.

04 / Constructed response

Which values in {11, 12, 13} satisfy x ≤ 12? Test each by substitution.

05 / Constructed response

Let x be any real number. Write and graph: x is more than 28. Label the endpoint and shade the solution ray.

06 / Constructed response

Which values in {7, 8, 9} satisfy x ≤ 8? Test each by substitution.

07 / Constructed response

Let x be any real number. Write and graph: x is at most 30. Give one included value and explain whether the endpoint is included.

08 / Multiple select

A club needs at least 28 members. Select all true statements for its membership count n.
A. 28 is an allowed membership count.
B. n ≥ 28
C. n > 28
D. 27 is an allowed membership count.

09 / Multiple select

A club needs at least 16 members. Select all true statements for its membership count n.
A. n ≥ 16
B. 15 is an allowed membership count.
C. n > 16
D. 16 is an allowed membership count.

10 / Constructed response

Which values in {4, 5, 6} satisfy x ≤ 5? Does the real-number inequality have only these solutions? Explain.

Day 070 | Unit 05 | Answer key

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1. x ≥ 21; closed circle at 21, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

2. 9, 10

Values below 10 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

3. x > 4; open circle at 4, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

4. 11, 12

Values below 12 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

5. x > 28; open circle at 28, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

6. 7, 8

Values below 8 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

7. x ≤ 30; closed circle at 30, shade left

“at most” means ≤; include the endpoint. Example solution: 29.

6.EE.B.8 | Constructed response

8. A, B - 28 is an allowed membership count.; n ≥ 28

At least includes the boundary, so n ≥ 28. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

9. A, D - n ≥ 16; 16 is an allowed membership count.

At least includes the boundary, so n ≥ 16. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

10. 4, 5; no, infinitely many real values satisfy it

Values below 5 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response